NISHANTH M

Cloud Infrastructure / DevOps Engineer | CI/CD, Automation & Observability | AI Infrastructure
+91 9489812743 | imnishanth23@gmail.com | linkedin.com/in/mnishanth23 | mnishanth23.github.io

PROFESSIONAL SUMMARY

Cloud Infrastructure and DevOps Engineer with 2+ years of hands-on experience building, automating and improving infrastructure for production applications, including AI workloads. Deployed and operate three vLLM model-serving containers on an NVIDIA A100 GPU and built GPU, container and host monitoring with DCGM exporter, cAdvisor, node-exporter, Prometheus and Grafana. Designed and maintain CI/CD pipelines in Jenkins and Drone CI with quality gates and release controls for consistently reliable production deployments, and automated provisioning end-to-end with Bash and Python, cutting environment setup from 2 days to 7–8 hours. Building a blue-green zero-downtime deployment platform on Azure using Terraform, Kubernetes (AKS), ACR and Argo Rollouts to deepen Kubernetes, IaC and GitOps-style progressive delivery. Primary responder for production incidents, troubleshooting across infrastructure and application layers. Curious about why tools are chosen, and proactively improves systems rather than only maintaining them.

TECHNICAL SKILLS

Kubernetes & Containers: Docker, Docker Compose (production); Kubernetes (Azure AKS), ACR and Argo Rollouts for blue-green delivery (personal project, in progress)

Infrastructure as Code: Terraform (personal project, in progress); Bash and Python Provisioning and Configuration Automation

CI/CD & Delivery: Jenkins, Drone CI, Jenkins-to-Drone Migration, Pipeline Quality Gates, Release Controls, Progressive Delivery Concepts, Git

Cloud: AWS (EC2, IAM, S3); Microsoft Azure AKS and ACR (personal project)

Monitoring & Tooling: Prometheus, Grafana, ELK Stack, OpenTelemetry, DCGM Exporter, cAdvisor, node-exporter, Alerting, Dashboards

AI Infrastructure: vLLM Model Serving on NVIDIA A100 (OCR models and quantised Llama 3.1 8B), GPU Memory Allocation and Telemetry

Linux & Scripting: Linux/VPS Administration, Python, Bash/Shell, SQL

Reliability & Security: Incident Response, Root-Cause Analysis, Runbooks, HashiCorp Vault, Least-Privilege Access, SSL/TLS Automation, SOC 2

Platforms & Data: HAProxy, Apache Karaf, ActiveMQ, PostgreSQL, MongoDB, Redis, ClickHouse

PROFESSIONAL EXPERIENCE

Associate DevOps Engineer | SMI Tech Group, Madurai, Tamil Nadu **Jan 2026 – Present**

- Deployed and operate three vLLM model-serving containers on a single NVIDIA A100 80GB GPU for a document-processing (IDP) platform, serving OCR and LLM models (LightOnOCR-2, Chandra OCR 2 and a quantised Llama 3.1 8B), and set up GPU and container monitoring with DCGM exporter, cAdvisor, node-exporter, Prometheus and Grafana
- Reduced full customer environment provisioning from 2 days to 7–8 hours by designing end-to-end Bash automation covering SSL certificate generation, Docker/Docker Compose, Karaf, Java and Python installation across development, test, staging and production tiers
- Delivered consistently reliable production deployments across customer-driven releases by enforcing pipeline quality gates, release controls and pre-production environment parity
- Cut database provisioning to under 5 minutes by scripting containerised deployment of PostgreSQL, MariaDB, MongoDB and Redis with least-privilege access configuration, eliminating manual setup
- Serve as primary responder for 1–2 production incidents per month, owning issues end-to-end through Level 2 support, structured root-cause analysis and documented fixes that prevent recurrence
- Migrated CI/CD tooling from Jenkins to Drone CI, improving pipeline maintainability and enabling faster, more reliable production releases
- Maintained SOC 2 audit readiness and auditability of production changes by owning compliance tracking and evidence collection in Sprinto

Junior DevOps Engineer | SMI Tech Group, Madurai, Tamil Nadu **Jun 2025 – Dec 2025**

- Designed and maintained CI/CD pipelines for containerised application deployments, reducing manual intervention in the release process
- Reduced repeat production incidents by leading root-cause analysis and post-incident reviews, converting findings into documented runbooks
- Improved environment consistency and reduced configuration drift across development and staging by supporting Docker-based deployment workflows
- Participated in on-call rotations and incident management, supporting platform stability and reliability objectives

Site Reliability Engineer (Trainee) | SMI Tech Group, Madurai, Tamil Nadu **Jan 2025 – May 2025**

- Improved on-call response by developing standardised runbooks and operational documentation for production environments, reducing time-to-acknowledge during incidents
- Strengthened platform stability by contributing operational improvements based on recurring incident patterns
- Maintained production environments in support of reliability and uptime objectives

Site Reliability Engineer (Intern) | SMI Tech Group, Madurai, Tamil Nadu **Jul 2024 – Dec 2024**

- Improved incident detection by implementing end-to-end monitoring, dashboards and alerting with Prometheus, Grafana and the ELK Stack across production infrastructure
- Reduced manual log review effort by building Python-based automation for log analysis and infrastructure operations
- Streamlined SOC 2 audit evidence collection by automating compliance workflows in Sprinto

PROJECTS

Blue-Green Zero-Downtime Deployment on Azure (In Progress) — Azure AKS, ACR, Terraform, Argo Rollouts, Prometheus, GitHub Actions

- Building a blue-green deployment platform on Azure, provisioning AKS and ACR with Terraform and using Argo Rollouts for zero-downtime releases, with planned Prometheus-based rollout analysis and GitHub Actions CI/CD

Web Phishing Detection Using Machine Learning — Python, TensorFlow, Scikit-learn, Flask

- Built a real-time phishing detection system with a Flask web interface for website legitimacy verification, integrated into browser and proxy environments

LSTM-Based Phishing Detection for Big E-mail Data — Python, TensorFlow, Keras

- Designed an LSTM deep-learning model for large-scale spam detection with scalable preprocessing pipelines; presented at ICRTSET 2K23 International Conference

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Engineering **2019 – 2023**

Loyola Institute of Technology and Science, Thovalai | CGPA: 7.8 / 10.0

CERTIFICATIONS & TRAINING

- Python Training — National Skill Development & Certification Authority, New Delhi
- Data Science Using Python Master Class — Scaler Academy
- Machine Learning Master Class — Pantech e-Learning
- Ethical Hacking Internship — Kaashiv Infotech, Chennai

LANGUAGES

English — Professional Proficiency | Tamil — Native Proficiency